

SDR Spectrum Sensing

Spectrum Sensing System Design

Selection: Choose a modular system aligned with SOSA (Sensor Open Systems Architecture) standards or similar open architectures. This ensures interoperability, allows for future upgrades, and avoids vendor lock-in.

USRP B210:

- Covers the desired frequency range (70 MHz to 6 GHz), including VHF (30-300 MHz) and C-band (4-8 GHz).
- Dual-channel operation allows for simultaneous monitoring or advanced signal processing.
- High-speed USB 3.0 ensures efficient data transfer for real-time applications.

Jetson Nano:

- GPU acceleration with 128 CUDA cores is ideal for parallel processing tasks like FFT (Fast Fourier Transform) for spectrum analysis.
- Supports machine learning frameworks (e.g., TensorFlow, PyTorch) for advanced signal classification or anomaly detection.
- Compact and power-efficient, making it suitable for portable or embedded systems.

Similar projects: several SDR systems and projects that target spectrum sensing in the VHF to C-band range, as this spectrum is widely used for telecommunications, broadcasting, and radar applications.

DARPA Spectrum Collaboration Challenge (SC2): Focused on developing cognitive radio systems for dynamic spectrum access to monitor and analyze frequencies for detecting primary users and optimizing spectrum usage.

Cognitive Radio Testbeds: Research institutions like the University of California, Berkeley, and Virginia Tech have developed SDR-based testbeds for spectrum sensing.

Commercial Solutions: Companies like ThinkRF and Per Vices offer SDR-based spectrum monitoring solutions that cover VHF to C-band.

Open-Source Projects:

- GitHub: GNU Radio
- OpenAirInterface (OAI): An open-source platform for 4G/5G experimentation, which includes spectrum sensing modules. GitHub: OpenAirInterface
- SigMF (Signal Metadata Format): A standard for storing and sharing SDR data, often used in spectrum monitoring projects.

- **Enhanced Real-Time Processing**
 Jetson Nano’s GPU for FFT computations with libraries like CuFFT.
 Minimize data transfer bottlenecks between USRP B210 and Jetson Nano by optimizing USB 3.0 throughput.
- **User Interface Development:**
 Develop a web-based UI using Flask or Django (Python) for remote access to spectrum data. Alternatively, use additional resources for a desktop application with real-time spectrum visualization:
 gr-speccet: GNU Radio spectral estimation toolkit
 LTE Cell Scanner: Frequency scanning implementations
 Pothos SDR: Alternative flow-based SDR framework

Antenna System Configurations

Option 1: Multi-Antenna System (Recommended)

Dual-Channel Advantage of USRP B210

- **Channel 1:** VHF/UHF Antenna (70 MHz – 3 GHz)
- **Channel 2:** C-Band Antenna (4 GHz – 6 GHz)

Antenna Selection

Band	Antenna Type	Model Example	Frequency Range	Gain
VHF/UHF	Discone	Comet DS-150B	25 MHz – 1.3 GHz	2–3 dBi
VHF/UHF	Log-Periodic	Aaronia HyperLOG 7060	700 MHz – 6 GHz	5–7 dBi
C-Band	Horn Antenna	Pasternack PE9851	4–8 GHz	10–15 dBi
C-Band	Helical	Custom/DIY	4–6 GHz	12–14 dBi

Table 1: Antenna Selection for Multi-Antenna System

Option 2: Wideband Compromise Antenna

Log-Periodic Dipole Array (LPDA)

Recommended Models:

- **Aaronia HyperLOG 7060 X:** 700 MHz – 6 GHz
- **Aaronia HyperLOG 30100:** 380 MHz – 10.5 GHz (covers full range)
- **A.H. Systems SAS-545:** 500 MHz – 6 GHz

Advantages:

- ✓ Single antenna solution
- ✓ Decent gain across wide bandwidth (4–6 dBi)
- ✓ Good directivity for interference rejection
- ✓ Commercially available

Disadvantages:

- △ Reduced sensitivity at band edges
- △ Directional (requires positioning/rotation)
- △ Larger physical size

Option 3: Automatic Antenna Switching System**Configuration**

```
USRP B210 <-- RF Switch <-- Multiple Band-Specific Antennas
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Control via Jetson Nano
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Components

RF Switch: Mini-Circuits USB/GPIO-controlled RF switch

- Example: **ZSWA-4-30DR+** (DC–3 GHz, 4-port)
- Example: **PE42442** (9 kHz – 4 GHz)

Antenna Array:

1. **70–500 MHz:** Discone or Dipole
2. **500 MHz – 2 GHz:** LPDA or Yagi
3. **2–4 GHz:** Patch array or Helical
4. **4–6 GHz:** Horn or C-band patch

Advantages:

- ✓ Optimized performance per band
- ✓ Automated frequency-based switching
- ✓ Maximum sensitivity across full range

Disadvantages:

- △ Increased system complexity
- △ Additional cost
- △ Switching time delays

Antenna System Budget Options

DIY Low-Cost Setup

Band	Antenna	Materials	Cost
VHF	Discone	Copper wire, PVC pipe	\$20–40
UHF-Low	Log-Periodic	Aluminum rod, boom	\$50–80
C-Band	Helical	Copper wire, PVC, N-connector	\$30–50
Total DIY			\$100–170
vs. Commercial			\$500–1,500

Supporting Hardware

Low Noise Amplifiers

Model	Range	Gain	NF
Nooelec Sawbird+	Multi-band	20 dB	<1 dB
Mini-Circuits ZX60-33LN-S+	50 MHz–3 GHz	20 dB	3 dB
SparkGap LNA4ALL	28 MHz–2.5 GHz	20 dB	1.5 dB

System Configurations

Budget Monitor (\$200–400): Discone + DIY Helical, manual switch, optional LNA

Professional (\$1,200–2,000): HyperLOG 30100, automated switching, per-band LNAs

Research-Grade (\$2,500–4,000): Multi-antenna array, automated 4-way switch, motorized rotator

Final Recommendation

The **dual-antenna system** offers the best balance:

- RX1: Aaronia HyperLOG (380 MHz–6 GHz+)
- RX2: Discone (70–300 MHz VHF)
- Cost: \$600–1,000
- No switching delays, simultaneous monitoring

Best Overall Solution: Dual-Antenna System

For spectrum sensing applications, the dual-antenna configuration provides optimal performance by fully leveraging the USRP B210's dual-channel capability.

Primary Setup

- **RX1 (USRP B210):** Aaronia HyperLOG 30100 or 7060
 - Coverage: 380 MHz – 6 GHz+ (full range)
- **RX2 (USRP B210):** Discone (Comet DS-150B)
 - Coverage: 70–300 MHz (VHF optimization)

Optional Enhancement

- Per-channel Low Noise Amplifiers (LNAs) for improved sensitivity

Advantages

- ✓ Leverages USRP B210's dual-channel capability
- ✓ Optimized coverage across full spectrum
- ✓ Simultaneous monitoring possible
- ✓ No switching delays
- ✓ Reasonable cost (\$600–1,000)
- ✓ Simple setup and maintenance

Mounting Considerations

Indoor Monitoring

For indoor spectrum monitoring installations, the following mounting options are recommended:

- **Desktop magnetic mount stands** – Provide stable antenna positioning on metal surfaces
- **Window-mounted antennas** – Optimize signal reception while minimizing indoor attenuation
- **Short quality cables (6 ft max)** – Minimize signal loss in short-range installations

Outdoor/Rooftop Monitoring

Outdoor installations require additional considerations for weather resistance and safety:

- **Non-penetrating roof mounts** – Avoid roof penetration while providing secure mounting
- **Mast mounting (1.25"–2" diameter)** – Standard mast sizes for antenna installation

- **Weatherproof antenna enclosures** – Protect sensitive RF components from environmental exposure
- **Grounding and lightning protection** – Essential safety measures for outdoor installations
- **Quality low-loss cables (LMR-400)** – Minimize signal degradation over longer cable runs

Testing & Calibration

Antenna Performance Verification

Proper system characterization requires several key measurements:

VSWR Measurement: Use NanoVNA or similar vector network analyzer to verify antenna impedance matching

Gain Pattern: Compare received signal levels against known reference signals to validate antenna gain

Frequency Response: Perform frequency sweep tests with USRP to characterize system response across bands

Noise Floor: Establish baseline noise measurements for sensitivity analysis

Software Tools

The following software tools are recommended for system testing and operation:

GNU Radio: Spectrum analyzer flowgraph development and signal processing

GQRX: Real-time spectrum visualization and monitoring

URH (Universal Radio Hacker): Advanced signal analysis and protocol decoding